CLASS 9 · CBSE · GANITA MANJARI PART 1 · FORMULA BANK

Chapter 5: I'm Up and Down, and Round and Round

Every formula and result of the chapter on one sheet

1	Circle	All points at distance r from the centre
2	Circles through 3 non-collinear points	Exactly one
3	Circumcentre	Meeting point of the perpendicular bisectors of the sides
4	Equal chords	Subtend equal angles at the centre, and conversely
5	Centre to mid-point of a chord	Perpendicular to the chord
6	Perpendicular from the centre	Bisects the chord
7	Length of a chord	$2\sqrt{r^2 - d^2}$, where d is its distance from the centre
8	Equal chords	Equidistant from the centre, and conversely
9	Unequal chords	The longer chord is nearer to the centre
10	Angle at the centre	$2\times$ angle at the remaining part of the circle
11	Angles in the same segment	Equal
12	Angle in a semicircle	90°
13	Cyclic quadrilateral	Opposite angles add up to 180°, and conversely
14	Exterior angle of a cyclic quadrilateral	Equal to the interior opposite angle